

Anubhav Kumar Srivastava

MSCA PhD Fellow, ICFO Spain

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Professional summary

MSCA PhD Fellow at ICFO – Institute of Photonic Sciences, Spain, specializing in quantum simulation, quantum metrology, error correction, and large-scale classical simulation of many-body systems.

7+ years of programming experience developing high-performance simulation tools, optimization pipelines, and machine-learning models for quantum systems. Strong background in tensor networks, semi-definite programming, and variational circuit optimization.

Seeking R&D and consulting roles in **quantum computing**, **quantum simulation**, and **quantum error correction**.

Education

Institute of Photonic Sciences (ICFO), Spain

Oct 2021 – May 2026 (expected)

Ph.D. in Quantum Physics, ENLIGHTEN Marie Skłodowska-Curie Co-fund Fellow

Supervisor: Prof. Dr. Maciej Lewenstein, **Co-supervisor:** Dr. Marcin Płodzień

Title: Quantum simulation of many-body spin systems for optimal quantum metrology

Indian Institute of Science Education & Research Mohali, India

Aug 2016 – June 2021

BS-MS (Dual) Degree, INSPIRE Scholarship for Higher Education

Major: Physics, **Minor:** Data Science, **CPI:** 8.6/10

MS thesis title: Resetting Quantum Systems through Superposition of Evolution, **Supervisor:** Dr. Manabendra Nath Bera

Technical Skills (Github link)

Programming: Python (advanced), Julia, Java, and C++

Quantum SDKs: Qiskit, PennyLane, QuSpin, QuTiP

Numerical & Optimization: Tensor networks (iTensor), semi-definite programming, convex optimization, quantum circuits

Machine learning: PyTorch, scikit-learn, attention-based neural networks, quantum machine learning

Scientific Computing: Simulation of many-body systems, variational algorithms, Hamiltonian learning

Research Experience

Los Alamos National Laboratory, USA

2024

Quantum Computing Summer School 2024: With Dr. A. Sornborger –

Thresholded magnetic field sensing with frustrated spin systems.

Indian Institute of Technology Bombay, India

2019

Internship: With Dr. S. Vinjanampathy –

Strongly interacting open quantum systems.

Indian Institute of Technology Kanpur, India

2018

Internship: With Dr. R. Mittal –

Basics of quantum algorithms (Grover's search and Shor's factorization).

Indian Institute of Technology Kanpur, India

2017

Internship: With Dr. S. Ghosh –

Designing PID control systems for coherent quantum optics laser setups.

Conferences & Talks

Invited Talks (1) : Quantum theory group (University of Geneva, 2024)

Contributed Talks (6): YQIS (Barcelona, 2025), FISMAT (Venice, 2025), ICFO-IMPRS workshop (Munich 2025, Barcelona 2023), ICE-9 (Tenerife, 2024), LANL QCSS (Los Alamos, 2024), YAO (Barcelona, 2023)

Poster presentations (8): XL Trobades (Mahón, 2025), Quantum matter out of equilibrium (Granada, 2025), CQD (Okinawa, 2024), QUMINOS (Les Diablerets, 2024), QIPA (Prayagraj, 2023), QTD (Vienna, 2023), Quantum simulation (Trieste, 2023), Quantum mixtures (Varenna, 2022)

Publications (Google Scholar link)

- Non-linear quantum sensing and frustrated Kitaev trimers, Alderete, C. H.*, **Srivastava, A. K.***, & Sornborger, A. T., *arXiv:2507.10418* (2025). (Accepted in PRR, Feb 2026)
- Topological quantum thermometry, **Srivastava, A. K.**, Bhattacharya, U., Lewenstein, M., & Płodzień, M., *Phys. Rev. A* 111, 052216 (2025).
- Introduction to quantum entanglement in many-body systems, **Srivastava, A. K.***, Müller-Rigat, G.*, Lewenstein, M., & Rajchel-Mieldzioć, G., *Lecture Notes in Physics, vol 1025. Springer, Cham* (2024).
- Enhancing quantum state tomography via resource-efficient attention-based neural networks, Palmieri, A. M., Müller-Rigat, G., **Srivastava, A. K.**, Lewenstein, M., Rajchel-Mieldzioć, G., & Płodzień, M., *Phys. Rev. Research* 6, 033248 (2024).
- Certifying the quantum Fisher information from a given set of mean values: a semidefinite programming approach, Müller-Rigat, G., **Srivastava, A. K.**, Kurdziałek, S., Rajchel-Mieldzioć, G., Lewenstein, M., & Frérot, I., *Quantum* 7, 1152 (2023).
- **Ongoing projects**: Classical shadows framework for quantum metrology and quantum thermometry, and ML tools for quantum state certification and modeling protein dynamics in chlorophyll of plants.

Leadership & Community

- **Frontiers for Young Minds (FYM)**: Accepted (2025) to contribute a research article, *Infinite multiverses and where to find them?*, to the special MSCA edition of FYM journal for children aged 8-16 in 230+ countries.
- **Co-supervised** two Bachelor's and Master's students in quantum simulation and ML-based biophysics projects (2025-26).
- **ICFO Young Photonics**: As part of the outreach program at ICFO, **taught and mentored** school students on the basics of quantum mechanics and introduction to quantum computing in 2023 and 2025.
- **Co-founder** (2021) of QTPi, a quantum awareness hub with over 300+ Discord members and 10+ YT videos as part of the global outreach quantum lecture series.
- **Turing club**: **Co-founder** and former **co-convener** of the coding club at IISER Mohali.
- **Group coordinator**: Organizing weekly group meetings and scientific guest visits for the **QOT group** at ICFO (2022-25).
- **Workshops**: Regularly participated in **soft-skills workshops** on **public speaking, research ethics, and scientific communication**, organized by ICFO and BIST, Spain.

Languages

English (Fluent), Spanish (Intermediate), French (Beginner), Hindi (Native speaker)

Other Information

Additional interests: Professionally trained in classical Indian music on guitar; active in quizzing and stand-up comedy; passionate about football and team sports.

Availability: Willing to relocate and travel internationally.

References: Available upon request.